



MPCA Project

TITLE OF THE PROJECT : Smart Automatic Hand Sanitizer
Dispenser using Arduino with Multi-Sensor Validation

Team Members:

Sl. No	Name	SRN
1	Vishal H	PES1UG24AM322
2	Aryaman DK	PES1UG24AM344

Components used:

- Arduino Uno
- Ultrasonic Sensor (HC-SR04)
- IR Sensor
- PIR Sensor
- Digital Temperature Sensor Module
- Servo Motor (GS5515MG)
- 16x2 LCD Display
- Breadboard
- Jumper Wires
- Laptop (Power Supply)

Explanation about the project :

This project implements a smart and touchless hand sanitizer dispensing system using Arduino. The system is designed to ensure hygiene and avoid false triggering by validating the presence of a real human hand using multiple sensors.

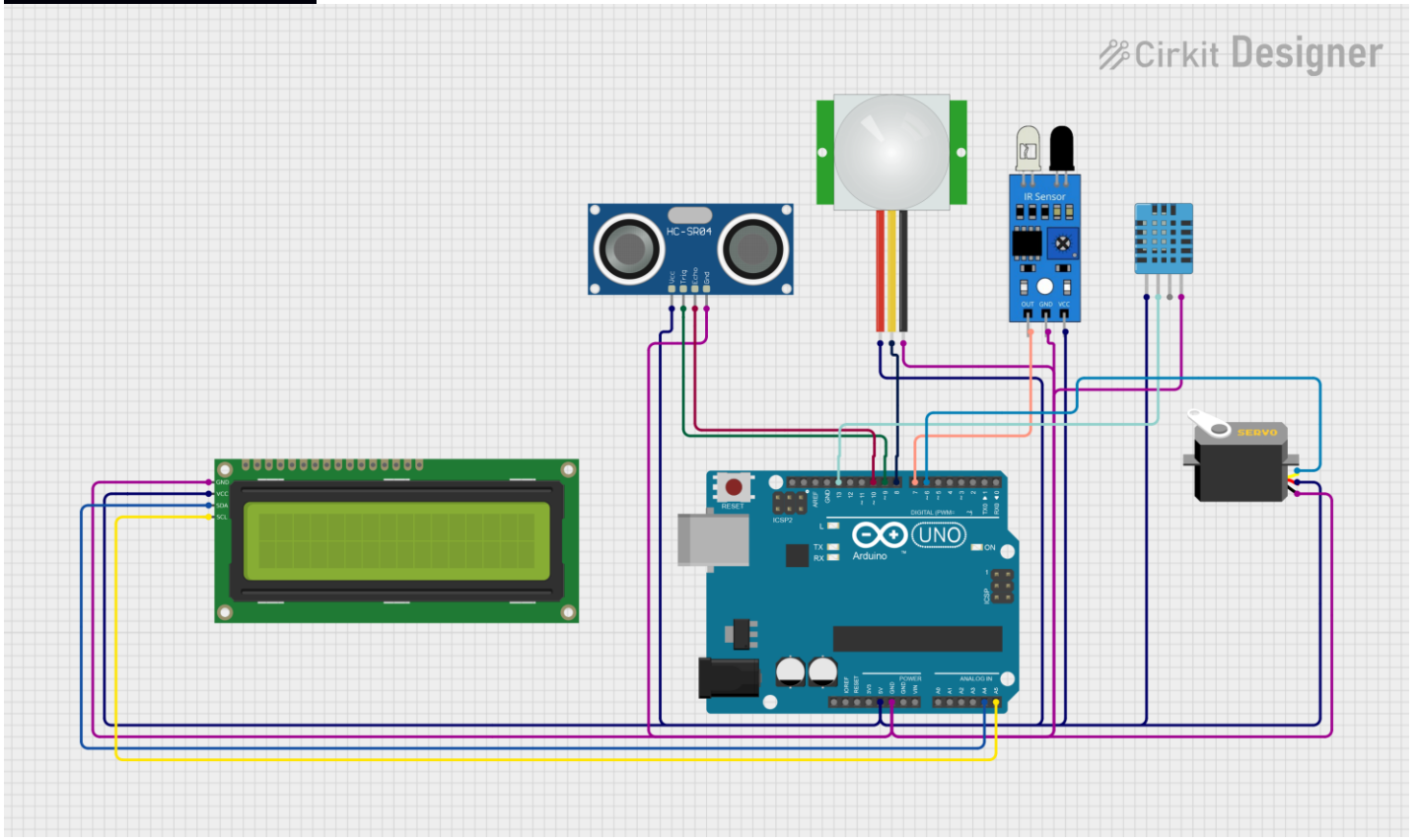
Initially, the ultrasonic sensor continuously measures the distance to detect the presence of an object. When an object is detected within a predefined range, the system begins the validation process.

The IR sensor confirms that the object is physically close, while the PIR sensor detects human motion based on infrared radiation. Additionally, the temperature sensor ensures that the object is warm, indicating a human hand.

Only when all these conditions are satisfied, the Arduino activates the servo motor, which presses the sanitizer pump. The LCD display provides real-time feedback to the user by showing messages such as system status, validation stages, and dispensing confirmation.

This multi-sensor approach ensures accurate detection and prevents unnecessary dispensing caused by non-human objects.

Circuit Diagram:



Pseudocode:

START

Initialize LCD, sensors, and servo

Display "Sanitizer Ready"

Display "Place Hand"

LOOP:

Measure distance using ultrasonic sensor

IF distance < threshold THEN
 Display "Detecting..."

Read IR sensor
 Read PIR sensor
 Read Temperature sensor

Display "Validating..."

IF (IR detects object AND PIR detects human AND Temperature is HIGH)
 THEN

Display "Dispensing..."
 Rotate servo to press pump
 Wait
 Reset servo
 Display "Thank You!"
 Delay to prevent repeat

ELSE

Display "Invalid Object"
 Display "Try Again"

ENDIF

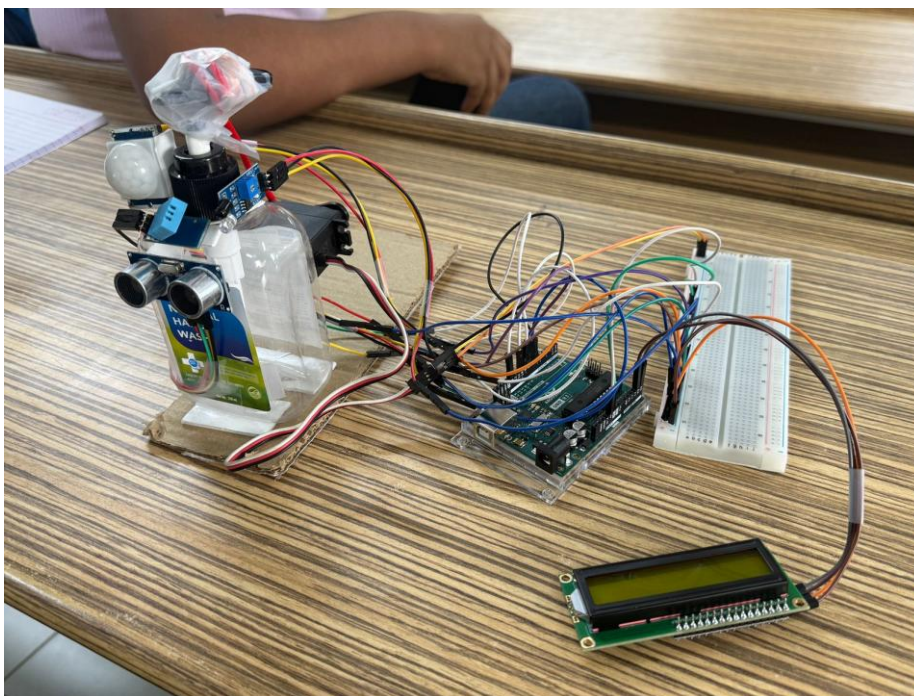
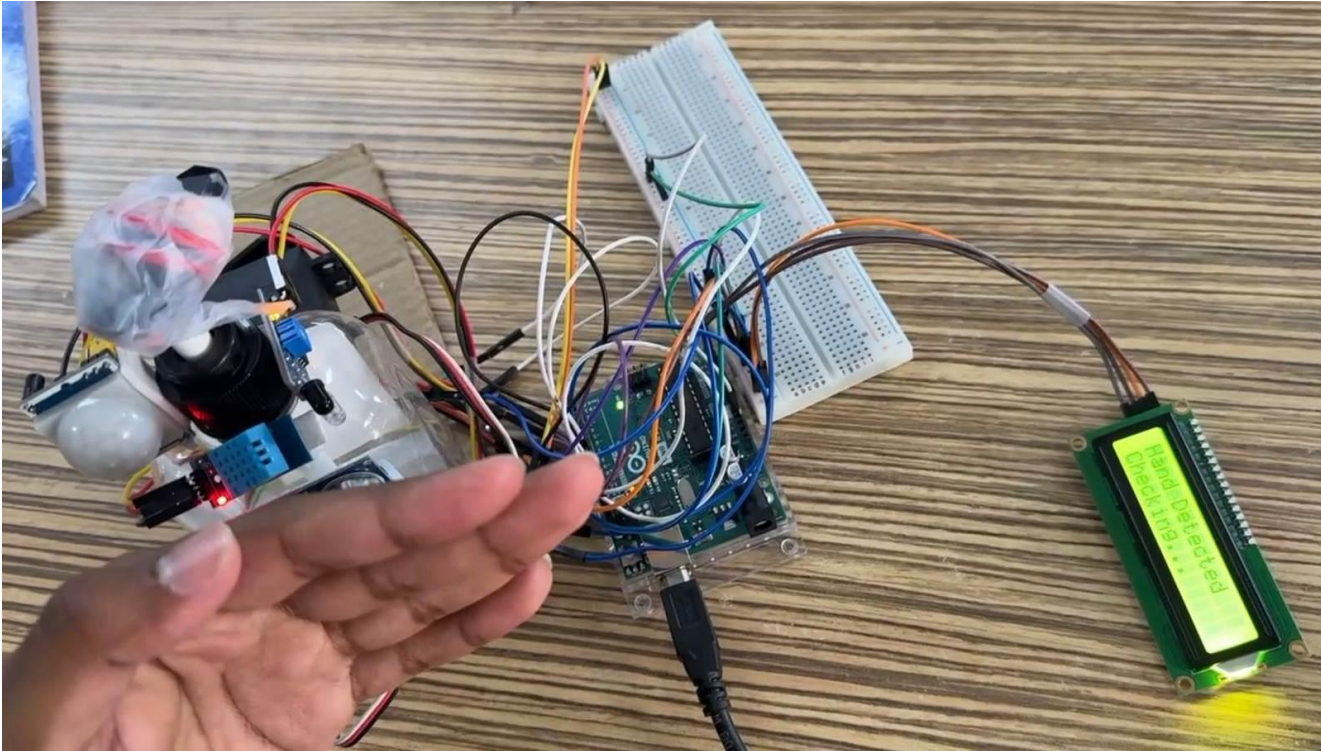
ELSE

Display "Place Hand"
ENDIF

END LOOP

END

Output Pictures with explanations:



- When the system is powered on, the LCD displays “Sanitizer Ready” and “Place Hand”, indicating that the system is active and waiting for user input.
- When a hand is placed below the sensor, the system detects it and displays “Detecting...” followed by “Validating...”.
- If all sensor conditions are satisfied (object is close, human motion detected, and warmth detected), the servo motor activates and presses the sanitizer pump. The LCD displays “Dispensing...” and then “Thank You!”.
- If a non-human object is placed or validation fails, the LCD shows “Invalid Object” and “Try Again”, and no sanitizer is dispensed.

Connections:

Component	Arduino Pin
LCD SDA	A4
LCD SCL	A5
Ultrasonic TRIG	9
Ultrasonic ECHO	10
IR Sensor	7
PIR Sensor	8
Temp Sensor (D)	13
Servo	6